Web Programming Using Python and JavaScript Lab

[SEC035]

Lab Report Submitted to

Bachelor of Technology

In

Computer Science Engineering

SEMESTER – 4th

Submitted by

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SECTION - B

Course Teacher

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School of Engineering & Technology

K. R. MANGALAM UNIVERSITY

EXPERIMENT NO – 1

AIM - To design and develop a personal portfolio website using HTML, CSS, and JavaScript that displays personal information, skills, projects, services, and contact details in a structured, attractive, and interactive manner.

Technologies Used -

- **HTML – For creating the structure of the website**
- **CSS – For styling, layout, colors, and responsiveness**
- **JavaScript – For adding interactivity like buttons, show/hide, and navigation effects**
- **Web Browser – To run and test the website**

Explanation –

1 HTML Structure

- **index.html contains the main structure of the portfolio.**
- **Semantic tags such as <header>, <nav>, <section>, and <footer> are used.**
- **Separate HTML pages are created for different services like frontend, database, and Java.**

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2 CSS Styling

- **style.css is linked externally to all HTML files.**
- **CSS is used for:**
  - **Layout alignment**
  - **Background colors and fonts**
  - **Spacing using margin and padding**
  - **Styling buttons, cards, and sections**

3️⃣ JavaScript Functionality

- **script.js is used to:**
  - **Handle button clicks**
  - **Add interactive effects**
  - **Improve navigation experience**
  - **Add show/hide functionality**

CODE-

```html
<!--
Project Title: Personal Portfolio Website
Name: Ananya Sharma
Date: 21 January 2026
-->

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Personal Portfolio</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>

    <!-- Header / Navigation -->
    <header id="header">
        <nav class="navbar">
            <h1 class="logo">Ananya Sharma</h1>
            <ul class="nav-links" id="navLinks">
                <li><a href="#about">About</a></li>
                <li><a href="#skills">Skills</a></li>
                <li><a href="#projects">Projects</a></li>
                <li><a href="#services">Services</a></li>
                <li><a href="#contact">Contact</a></li>
            </ul>
        </nav>

        <div class="hero">
            <h2 id="welcomeText">Hello, I'm Ananya 👋</h2>
            <p>Creative Student | Web Development Enthusiast</p>
            <a href="#projects" class="btn" id="viewWorkBtn">View My Work</a>
        </div>
    </header>
```

```html
<!-- About Me Section -->
<section id="about" class="section">
    <h2>About Me</h2>
    <p>
        Hello! My name is Ananya Sharma, and I am a passionate and motivated student
        with a strong interest in web development and emerging technologies.
        I enjoy designing clean, user-friendly websites and turning ideas into
        functional digital solutions. I have hands-on experience in building
        projects using HTML and CSS, and I am continuously improving my skills
        in modern web technologies.
    </p>
</section>

<!-- Skills Section -->
<section id="skills" class="section">
    <h2>Skills</h2>
    <ul class="skills-list">
        <li>HTML5</li>
        <li>CSS3</li>
        <li>Basic JavaScript</li>
        <li>SQL & DBMS Basics</li>
        <li>Problem Solving</li>
        <li>Communication Skills</li>
    </ul>
</section>

<!-- Projects Section -->
<section id="projects" class="section">
    <h2>Projects</h2>

    <div class="project-card">
        <h3>Instock - Inventory Management Website</h3>
        <p>
            A responsive inventory management website developed using HTML and CSS.
            This project helps in organizing and displaying stock information in a
            clean and structured layout.
        </p>
    </div>

    <div class="project-card">
        <h3>Recipe Haven - Recipe Sharing Website</h3>
        <p>
            A visually appealing recipe website built using HTML and CSS where users
            can explore different recipes with attractive layouts and easy navigation.
        </p>
    </div>
</section>

<!-- Services Section -->
<section id="services" class="section light-bg">
    <h2>Services</h2>
    <div class="services-grid">
        <div class="service-card" onclick="highlightService(this)">
```

```html
            I create responsive and visually appealing web pages using HTML, CSS,
            and basic JavaScript. The focus is on clean design, proper layout,
            and a user-friendly experience that works on all devices.
        </p>
        <a href="frontend.html" class="service-btn" target="_blank">Learn More</a>
    </div>

    <div class="service-card" onclick="highlightService(this)">
        <h3>Database Management</h3>
        <p>
            I work with SQL and DBMS to design, manage, and query databases efficiently.
            This includes creating tables, managing records, and ensuring data integrity
            for small to medium-scale applications.
        </p>
        <a href="database.html" class="service-btn" target="_blank">Learn More</a>
    </div>

    <div class="service-card" onclick="highlightService(this)">
        <h3>Java Programming</h3>
        <p>
            I have experience in writing Java programs for problem-solving,
            building basic applications, and implementing object-oriented concepts.
            I can develop functional and structured code for academic projects.
        </p>
        <a href="java.html" class="service-btn" target="_blank">Learn More</a>
    </div>
</div>
ection>


- Contact Section -->
ction id="contact" class="section contact-section">
>Contact Me</h2>
class="contact-text">


rm id="contactForm" class="contact-form">
<div class="form-group">
    <input type="text" id="name" placeholder="Your Full Name" required>
</div>

<div class="form-group">
    <input type="email" id="email" placeholder="Your Email Address" required>
</div>

<div class="form-group">
    <textarea id="message" placeholder="Your Message" rows="5" required></textarea>
</div>

<button type="submit" class="btn">Send Message</button>
```

CSS CODE

```
/* General Styles */
body {
    margin: 0;
    font-family: Arial, Helvetica, sans-serif;
    background-color: #f4f6f8;
    color: #333;
    line-height: 1.6;
}

/* Header & Navigation */
#header {
    background: linear-gradient(to right, #0f2027, #203a43, #2c5364);
    color: white;
    padding-bottom: 40px;
}

.navbar {
    display: flex;
    justify-content: space-between;
    align-items: center;
    padding: 20px 40px;
}

.logo {
    font-size: 28px;
    font-weight: bold;
}

.nav-links {
    list-style: none;
    display: flex;
    gap: 20px;
}

.nav-links li a {
    text-decoration: none;
    color: white;
    font-size: 16px;
}

.nav-links li a:hover {
    color: #ffcc00;
}

/* Hero Section */
.hero {
    text-align: center;
    margin-top: 40px;
}

.hero h2 {
    font-size: 36px;
    margin-bottom: 10px;
}
```

```
.hero p {
    font-size: 18px;
    margin-bottom: 20px;
}

.hero button {
    padding: 8px 18px;
    font-size: 14px;
    background-color: #ffcc00;
    border: none;
    border-radius: 6px;
    cursor: pointer;
    font-weight: bold;
}


.btn:hover {
    background-color: #e6b800;
}

/* Common Section Styling */
.section {
    padding: 60px 80px;
    background-color: white;
    margin: 30px auto;
    max-width: 1000px;
    border-radius: 10px;
    box-shadow: 0 0 10px rgba(0, 0, 0, 0.1);
}

.section h2 {
    text-align: center;
    margin-bottom: 30px;
    color: #2c5364;
}

.light-bg {
    background-color: #f9fafb;
}

/* About Section */
#about p {
    max-width: 750px;
    margin: auto;
    text-align: center;
    font-size: 17px;
}

/* Skills Section */
.skills-list {
    list-style: none;
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(150px, 1fr));
    gap: 15px;
    padding: 0;
```

```css
/* Projects Section */
.project-card {
    background-color: #f9f9f9;
    border-left: 5px solid #2c5364;
    padding: 20px;
    margin-bottom: 20px;
    border-radius: 6px;
}

.project-card h3 {
    margin-top: 0;
    color: #2c5364;
}

/* Services Section */
.services-grid {
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(250px, 1fr));
    gap: 20px;
}

.service-card {
    background-color: #f9f9f9;
    padding: 20px;
    border-radius: 6px;
    text-align: left;
    box-shadow: 0 4px 12px rgba(0,0,0,0.1);
}

.service-card h3 {
    margin-bottom: 12px;
    color: #2c5364;
}

.service-card p {
    font-size: 15px;
    line-height: 1.5;
    margin-bottom: 15px;
}

.service-btn {
    display: inline-block;
    padding: 8px 16px;
    background-color: #2563eb;
    color: white;
    border-radius: 5px;
    text-decoration: none;
    font-weight: bold;
}

.service-btn:hover {
    background-color: #1d4ed8;
}
```

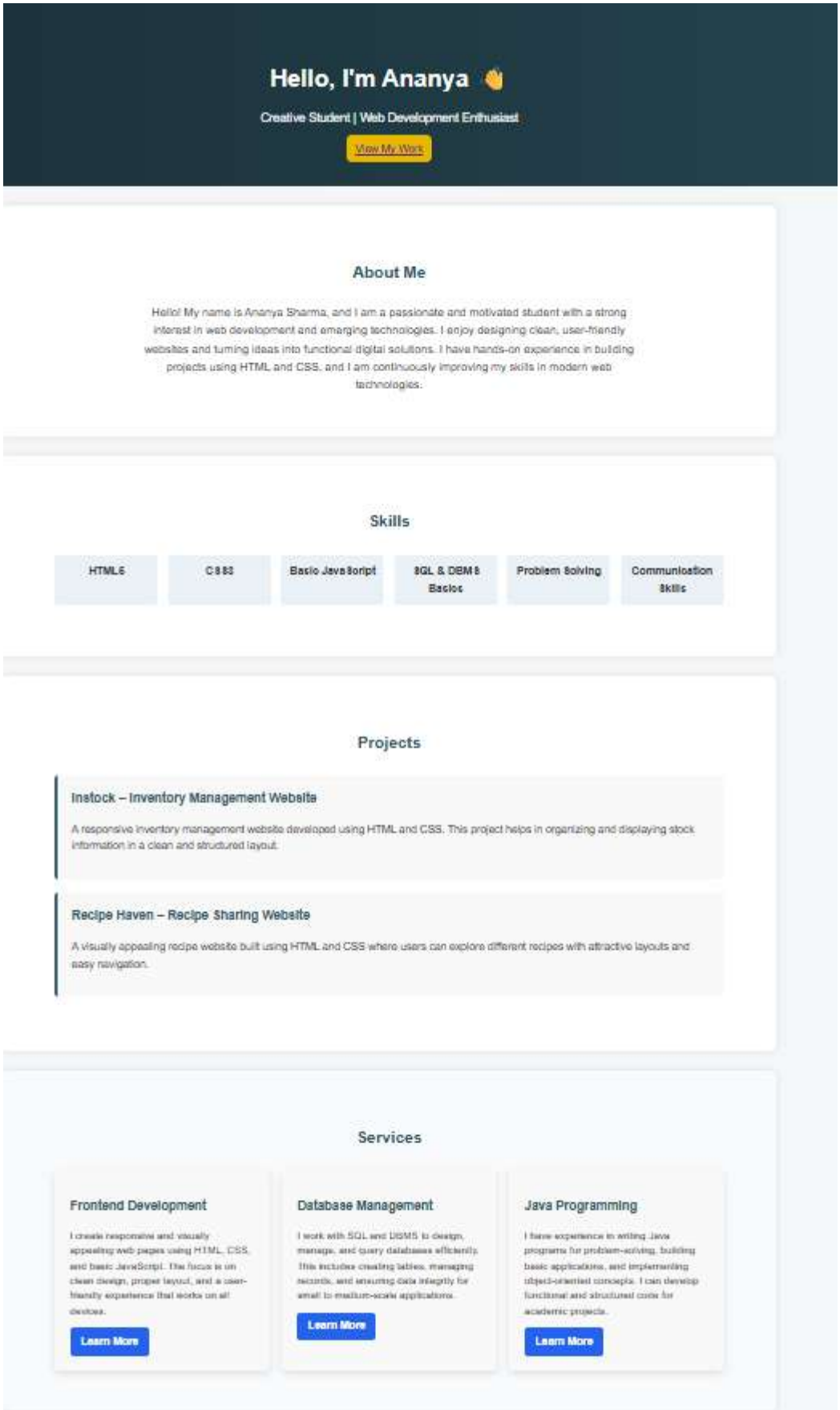
```
#contact p {
    text-align: center;
    font-size: 16px;
    margin: 8px 0;
}

/* Footer */
.footer {
    background-color: #0f2027;
    color: white;
    text-align: center;
    padding: 15px 0;
    margin-top: 40px;
}
.active-service {
    border: 2px solid #4f46e5;
    transform: scale(1.03);
    transition: 0.3s ease;
}
.service-card {
    border: 2px solid transparent;
    transition: 0.3s ease;
    cursor: pointer;
}

.service-card:hover {
    border: 2px solid #4f46e5;
}

.active-service {
    border: 2px solid #4f46e5 !important;
    box-shadow: 0 0 10px rgba(79, 70, 229, 0.3);
    transform: scale(1.03);
}
.contact-section {
    text-align: center;
    padding: 60px 20px;
}

.contact-text {
    max-width: 450px;
    margin: 0 auto 30px;
    text-align: center;
    line-height: 1.6;
}


.contact-form {
    max-width: 450px;
    margin: 0 auto;
    background: #f9fafb;
    padding: 30px;
    border-radius: 12px;
    box-shadow: 0 8px 20px rgba(0,0,0,0.05);
}
```

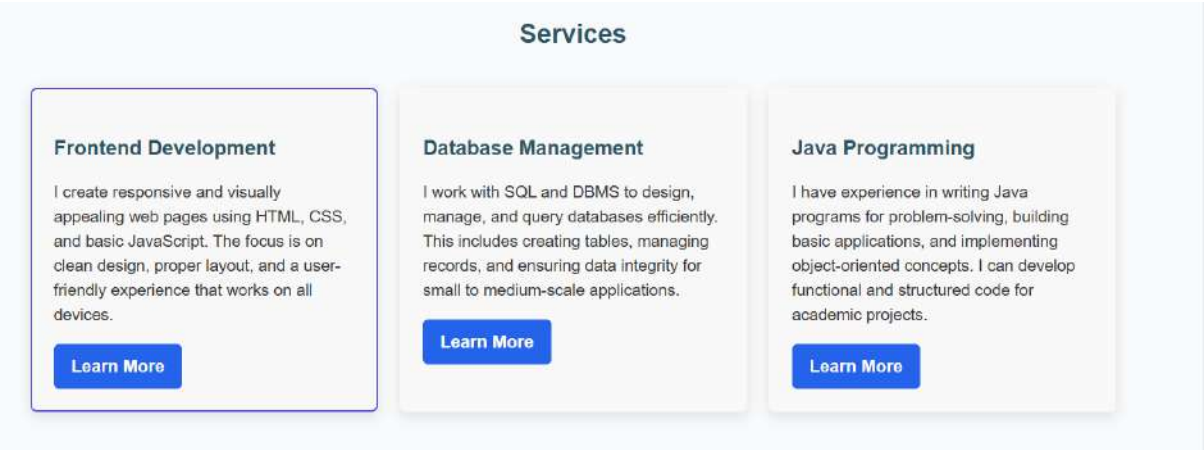
```css
.form-group {
    margin-bottom: 18px;
}

.contact-form input,
.contact-form textarea {
    width: 100%;
    padding: 12px 14px;
    border-radius: 8px;
    border: 1px solid #ccc;
    font-size: 15px;
    outline: none;
}

.contact-form input:focus,
.contact-form textarea:focus {
    border-color: #4f46e5;
    box-shadow: 0 0 0 2px rgba(79,70,229,0.1);
}

.btn {
    width: 100%;
    padding: 12px;
    font-size: 16px;
    border-radius: 8px;
}

.form-message {
    margin-top: 15px;
    font-weight: 600;
}
```

JavaScript Code-

```javascript
document.getElementById("viewWorkBtn").addEventListener("click", function (e) {
    e.preventDefault();
    document.getElementById("projects").scrollIntoView({ behavior: "smooth" });
});


function highlightService(card) {
    const cards = document.querySelectorAll(".service-card");
    cards.forEach(c => c.classList.remove("active-service"));
    card.classList.add("active-service");
}
```

OUTPUT


Hello, I'm Ananya 👋
Creative Student | Web Development Enthusiast
View My Work
About Me
Hello! My name is Ananya Sharma, and I am a passionate and motivated student with a strong interest in web development and emerging technologies. I enjoy designing clean, user-friendly websites and turning ideas into functional digital solutions. I have hands-on experience in building projects using HTML and CSS, and I am continuously improving my skills in modern web technologies.
Skills
HTML5
CSS3
Basic JavaScript
SQL & DBMS Basics
Problem Solving
Communication Skills
Projects
Instock – Inventory Management Website
A responsive inventory management website developed using HTML and CSS. This project helps in organizing and displaying stock information in a clean and structured layout.
Recipe Haven – Recipe Sharing Website
A visually appealing recipe website built using HTML and CSS where users can explore different recipes with attractive layouts and easy navigation.
Services
Frontend Development
I create responsive and visually appealing web pages using HTML, CSS, and basic JavaScript. The focus is on clean design, proper layout, and a user-friendly experience that works on all devices.
Learn More
Database Management
I work with SQL and DBMS to design, manage, and query databases efficiently. This includes creating tables, managing records, and ensuring data integrity for small to medium-scale applications.
Learn More
Java Programming
I have experience in writing Java programs for problem-solving, building basic applications, and implementing object-oriented concepts. I can develop functional and structured code for academic projects.
Learn More


Services
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I create responsive and visually appealing web pages using HTML, CSS, and basic JavaScript. The focus is on clean design, proper layout, and a user-friendly experience that works on all devices.
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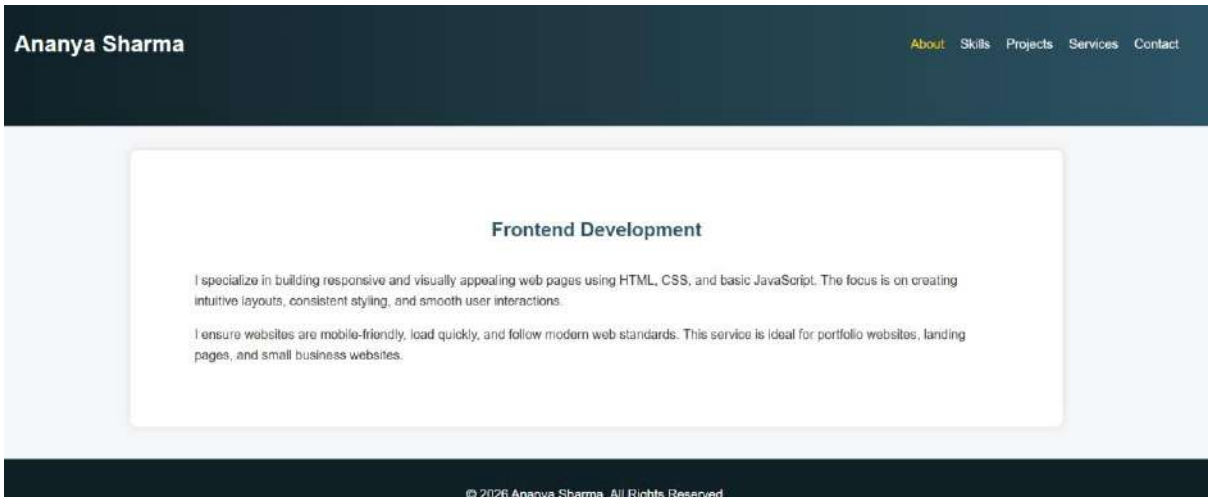
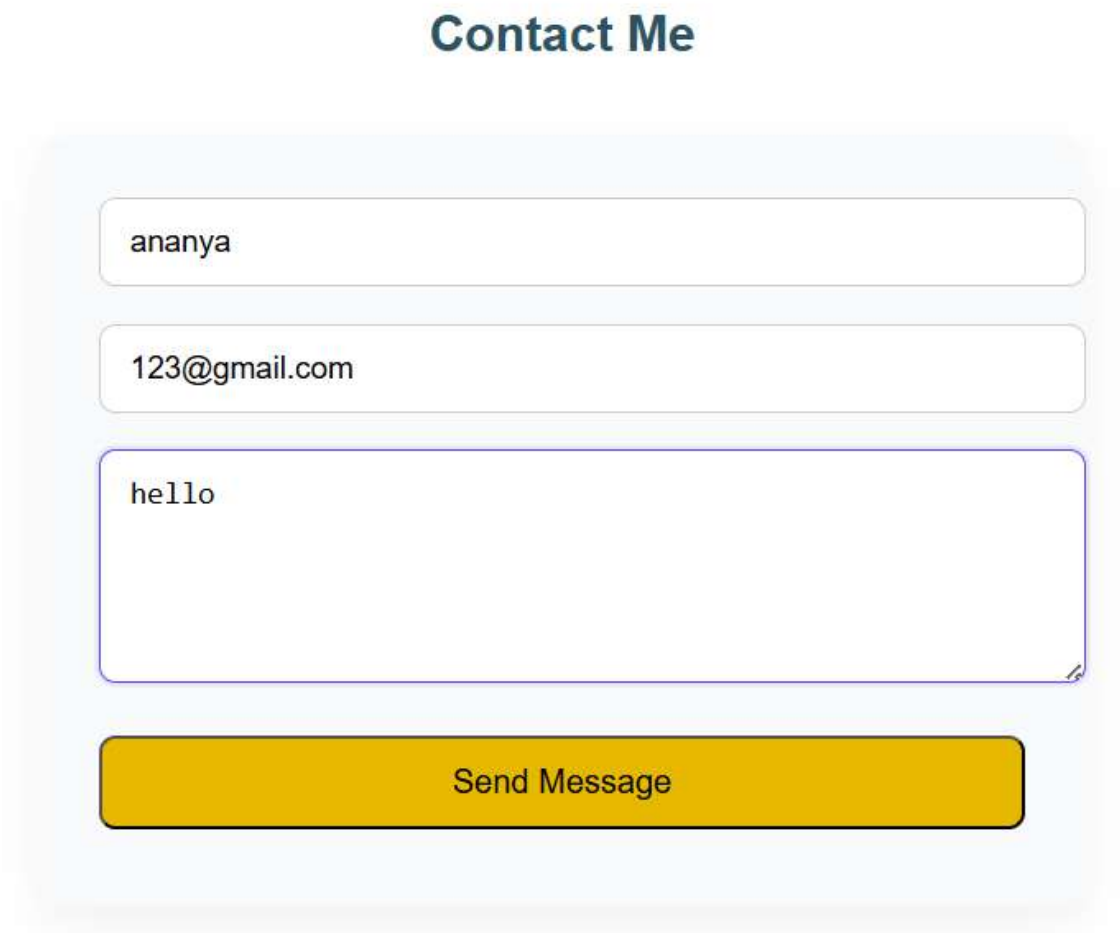
Learn More
Java Programming
I have experience in writing Java programs for problem-solving, building basic applications, and implementing object-oriented concepts. I can develop functional and structured code for academic projects.
Learn More


Ananya Sharma
About Skills Projects Services Contact
Frontend Development
I specialize in building responsive and visually appealing web pages using HTML, CSS, and basic JavaScript. The focus is on creating intuitive layouts, consistent styling, and smooth user interactions.
I ensure websites are mobile-friendly, load quickly, and follow modern web standards. This service is ideal for portfolio websites, landing pages, and small business websites.
© 2026 Ananya Sharma. All Rights Reserved



✅ **Result:**

A fully functional, professional-looking personal portfolio website was successfully created using HTML, CSS, and JavaScript.

EXPERIMENT NO – 2

AIM- To design and develop a professional company landing page named Bright Solution using HTML and CSS, displaying company information, services, testimonials, and contact details in a well-structured and visually appealing manner.

Technologies Used-

- HTML – To create the structure of the landing page

- CSS – To design layout, colors, fonts, and overall appearance

- Web Browser – To test and run the website

Explanation-

1️⃣ HTML Structure

- **index.html contains the main structure of the landing page.**
- **Semantic tags like <header>, <section>, and <footer> are used.**
- **Sections included:**
  - **Header / Hero Section**
  - **About / Introduction**
  - **Services**
  - **Testimonials**
  - **Contact Information**
- **Separate HTML pages are created for each service (Web Development, Database, IT Consulting).**

2️⃣ CSS Styling

- **style.css is linked externally to all pages.**
- **CSS is used to:**
  - **Style layout and spacing**
  - **Apply background colors and fonts**
  - **Align sections and create service cards**
  - **Improve readability and visual appeal**

CODE

HTML

```html
<!--
Project Title: Company Landing Page
Name: Ananya Sharma
Date: 2026
-->

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>BrightTech Solutions</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>

    <!-- Header / Hero Section -->
    <header class="hero">
        <div class="hero-content">
            <h1>BrightTech Solutions</h1>
            <p>Innovating Your Digital Future</p>
            <a href="#services" class="btn">Explore Our Services</a>
        </div>
    </header>

    <!-- About / Introduction Section -->
    <section id="about" class="section">
        <h2>About Us</h2>
        <p>
            BrightTech Solutions is a forward-thinking technology company dedicated to
            delivering high-quality digital solutions. We specialize in building modern,
            secure, and user-friendly applications that help businesses grow and succeed
            in the digital world.
        </p>
    </section>
```

```html
<body>

    <!-- Services Section -->
    <section id="services" class="section light-bg">
    <h2>Our Services</h2>
    <div class="services-grid">

        <a href="web-development.html" class="service-card-link">
            <div class="service-card">
                <h3>Web Development</h3>
                <p>
                    We design and develop responsive, fast, and visually appealing websites
                    tailored to meet your business needs and enhance user experience.
                </p>
            </div>
        </a>

        <a href="database-management.html" class="service-card-link">
            <div class="service-card">
                <h3>Database Management</h3>
                <p>
                    Our team provides efficient database design, management, and optimization
                    services to ensure secure and reliable data storage solutions.
                </p>
            </div>
        </a>

        <a href="it-consulting.html" class="service-card-link">
            <div class="service-card">
                <h3>IT Consulting</h3>
                <p>
                    We offer expert IT consulting services to help businesses adopt the right
                    technologies and improve their overall operational efficiency.
                </p>
            </div>
        </a>

    </div>
</section>


    <!-- Testimonials Section -->
    <section id="testimonials" class="section">
        <h2>What Our Clients Say</h2>
        <div class="testimonials-container">

            <div class="testimonial-card">
                <p>
                    "BrightTech Solutions transformed our online presence. Their team is
                    professional, responsive, and highly skilled. We are extremely satisfied
                    with their work!"
                </p>
                <h4>- Rahul Verma</h4>
            </div>
```

```html
<p>
    "The quality of service and attention to detail provided by BrightTech
    Solutions is outstanding. They delivered our project on time and exceeded
    our expectations."
</p>
<h4>- Neha Sharma</h4>
div>
```

```html
<!-- Contact Section -->
<section id="contact" class="section light-bg">
    <h2>Contact Us</h2>

    <div class="contact-info">
        <p><strong>Address:</strong> 123 Tech Street, New Delhi, India</p>
        <p><strong>Email:</strong> contact@brighttechsolutions.com</p>
        <p><strong>Phone:</strong> +91 98765 43210</p>
    </div>

    <form class="contact-form">
        <input type="text" placeholder="Your Name" required>
        <input type="email" placeholder="Your Email" required>
        <textarea rows="5" placeholder="Your Message" required></textarea>
        <button type="submit" class="btn">Send Message</button>
    </form>
</section>

<!-- Footer -->
<footer class="footer">
    <p>Designed by Ananya Sharma</p>
</footer>

/body>
/html>
```

CSS CODE

yle.css > ...

```
* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
    font-family: "Segoe UI", Tahoma, sans-serif;
}

body {
    background-color: #f4f7fb;
    color: #333;
    line-height: 1.6;
}

/* Hero Section */
.hero {
    background: linear-gradient(135deg, #0f2d3a, #2563eb);
    color: white;
    padding: 110px 20px;
    text-align: center;
}

.hero h1 {
    font-size: 50px;
    margin-bottom: 12px;
    font-weight: 700;
}

.hero p {
    font-size: 20px;
    margin-bottom: 28px;
    opacity: 0.9;
}

.btn {
    display: inline-block;
    padding: 12px 26px;
    background-color: #ffcc00;
    color: #000;
    text-decoration: none;
    border-radius: 30px;
    font-weight: 600;
    transition: 0.3s ease;
}

.btn:hover {
    background-color: #ffd633;
    transform: translateY(-2px);
}

/* Sections */
.section {
    padding: 80px 20px;
    max-width: 1150px;
    margin: auto;
    text-align: center;
```

```
    position: relative;
}

.section h2::after {
    content: "";
    width: 60px;
    height: 3px;
    background: #2563eb;
    display: block;
    margin: 10px auto 0;
    border-radius: 2px;
}

.light-bg {
    background-color: #eef3fb;
}

/* Services Section */
.services-grid {
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(260px, 1fr));
    gap: 25px;
}

.service-card {
    background: white;
    padding: 28px;
    border-radius: 14px;
    box-shadow: 0 10px 25px rgba(0,0,0,0.08);
    text-align: left;
    transition: 0.3s ease;
}

.service-card:hover {
    transform: translateY(-5px);
    box-shadow: 0 14px 35px rgba(0,0,0,0.12);
}

.service-card h3 {
    margin-bottom: 12px;
    color: #0f2d3a;
    font-size: 20px;
}

/* Testimonials Section */
.testimonials-container {
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(280px, 1fr));
    gap: 25px;
}
```

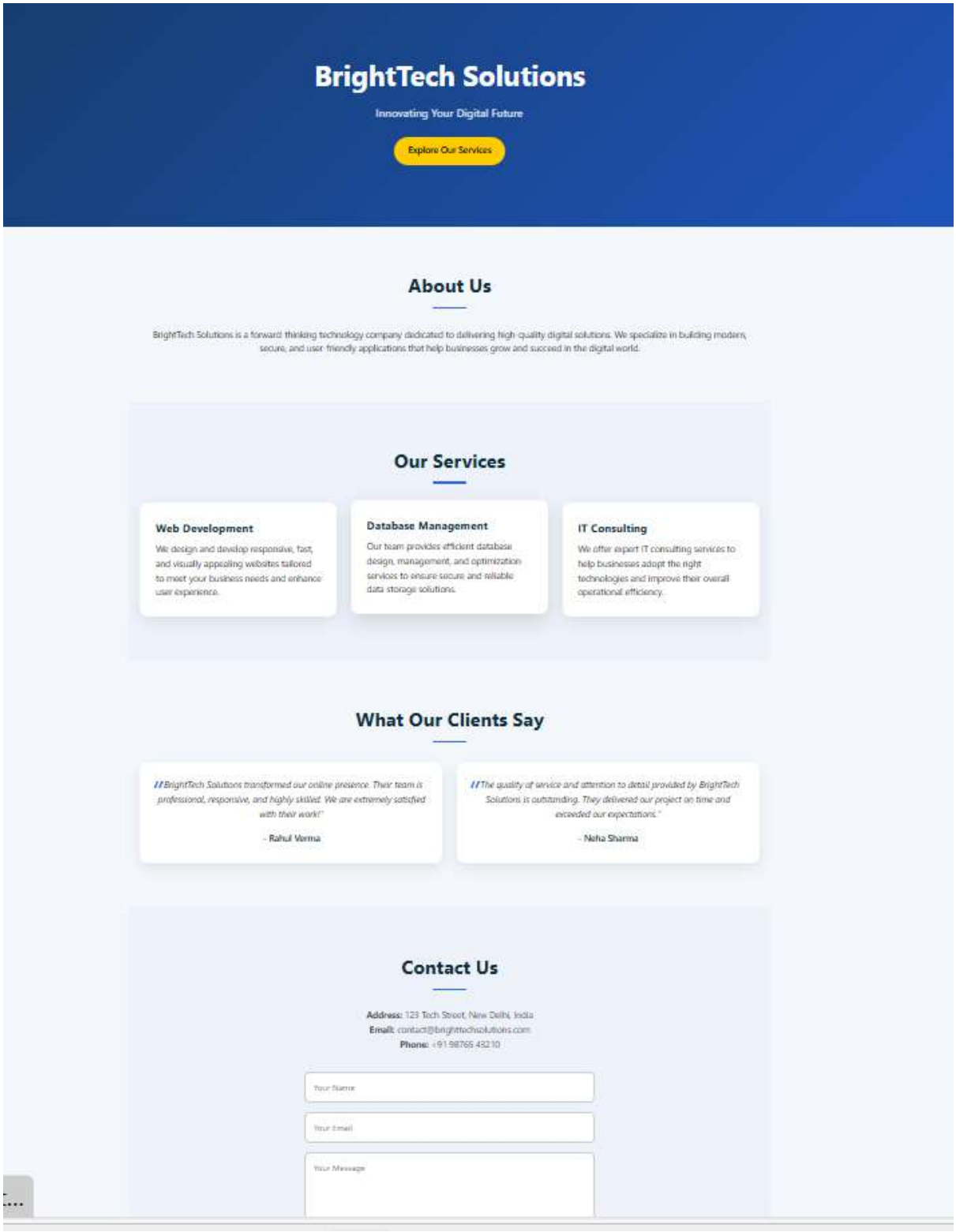
```
.testimonial-card {
    background: white;
    padding: 30px;
    border-radius: 14px;
    box-shadow: 0 10px 25px rgba(0,0,0,0.08);
    font-style: italic;
    position: relative;
}

.testimonial-card::before {
    content: "“";
    font-size: 50px;
    color: #2563eb;
    position: absolute;
    top: 10px;
    left: 20px;
}

.testimonial-card h4 {
    margin-top: 18px;
    color: #0f2d3a;
    font-style: normal;
    font-weight: 600;
}

/* Contact Section */
.contact-info {
    margin-bottom: 35px;
    font-size: 16px;
    color: #555;
}

.contact-form {
    max-width: 520px;
    margin: auto;
    display: flex;
    flex-direction: column;
    gap: 18px;
}

.contact-form input,
.contact-form textarea {
    padding: 14px;
    border-radius: 10px;
    border: 1px solid #ccc;
    font-size: 15px;
    transition: 0.2s ease;
```

```css
.contact-form input:focus,
.contact-form textarea:focus {
    border-color: #2563eb;
    outline: none;
    box-shadow: 0 0 0 3px rgba(37,99,235,0.1);
}

.contact-form button {
    padding: 14px;
    font-size: 16px;
    border: none;
    border-radius: 30px;
    background-color: #ffcc00;
    font-weight: 600;
    cursor: pointer;
    transition: 0.3s ease;
}

.contact-form button:hover {
    background-color: #ffd633;
    transform: translateY(-2px);
}

/* Footer */
.footer {
    background: #0f2d3a;
    color: white;
    text-align: center;
    padding: 18px;
    margin-top: 50px;
    font-size: 15px;
}
.service-card-link {
    text-decoration: none;
    color: inherit;
    display: block;
}
```

📌 **Features Implemented:**

- **Company name and tagline displayed in hero section**
- **Services displayed in card layout with clickable links**
- **Testimonials section with customer feedback**
- **Contact section with address, email, phone, and form**
- **Clean and professional UI**

OUTPUT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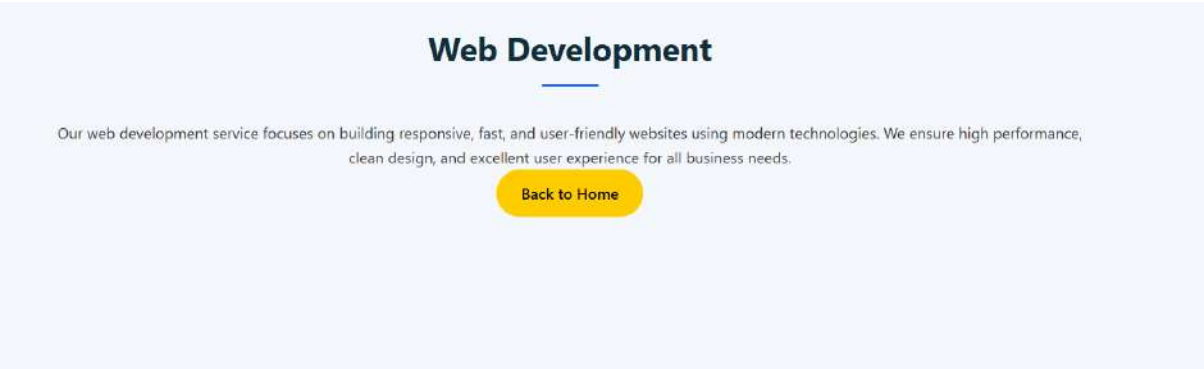
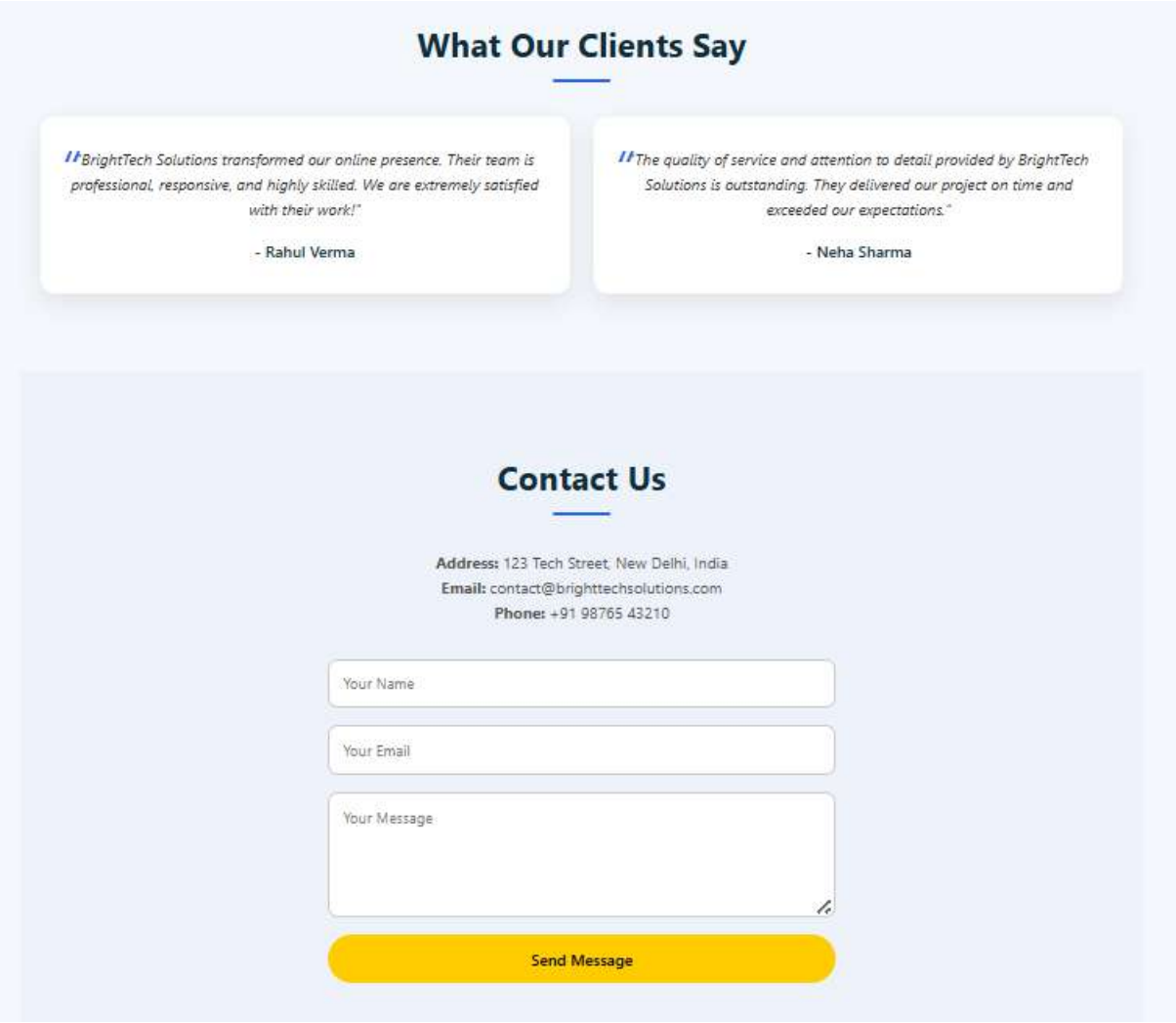
Innovating Your Digital Future
Explore Our Services
About Us
BrightTech Solutions is a forward thinking technology company dedicated to delivering high-quality digital solutions. We specialize in building modern, secure, and user-friendly applications that help businesses grow and succeed in the digital world.
Our Services
Web Development
We design and develop responsive, fast, and visually appealing websites tailored to meet your business needs and enhance user experience.
Database Management
Our team provides efficient database design, management, and optimization services to ensure secure and reliable data storage solutions.
IT Consulting
We offer expert IT consulting services to help businesses adopt the right technologies and improve their overall operational efficiency.
What Our Clients Say
"BrightTech Solutions transformed our online presence. Their team is professional, responsive, and highly skilled. We are extremely satisfied with their work!"
- Rahul Verma
"The quality of service and attention to detail provided by BrightTech Solutions is outstanding. They delivered our project on time and exceeded our expectations."
- Neha Sharma
Contact Us
Address: 123 Tech Street, New Delhi, India
Email: contact@brighttechsolutions.com
Phone: +91 98765 43210
Your Name
Your Email
Your Message





✅ **Result: A complete and visually appealing company landing page for Bright Solution was successfully developed using HTML and CSS.**

EXPERIMENT NO – 3

AIM- To design and develop an interactive quiz application using HTML, CSS, and JavaScript that displays questions, records user responses, tracks the score, shows results, and allows the quiz to be restarted.

Technologies Used:

- **HTML – For creating the structure of the quiz interface**
- **CSS – For styling the quiz layout and improving user experience**
- **JavaScript – For quiz logic, interactivity, timer, and result calculation**
- **Web Browser – To run and test the application**

Explanation

1 HTML Structure

HTML is used to create the layout of the quiz using semantic elements such as <header>, <section>, and <button>.
It displays:

- **Quiz title and introduction**
- **Question text**
- **Multiple-choice options (radio buttons)**
- **Next button**
- **Result section and feedback form**

2️⃣ CSS Styling

CSS is applied to:

- **Improve readability of questions and options**
- **Align sections and buttons properly**
- **Use colors, spacing, and borders for better user experience**
- **Make the quiz interface clean and professional**

3️⃣ JavaScript Functionality

JavaScript is used to:

- **Store questions and answers in an array of objects**
- **Capture user-selected answers**
- **Move between questions using button events**
- **Track and update the score**
- **Display final results and feedback messages**
- **Implement a countdown timer for each question**
- **Allow the quiz to restart**

CODE

HTML CODE

script.js index.html

index.html > html > body

```html
<!--
Project Title: Interactive Quiz Application
Name: Ananya Sharma
Date: 21 January 2026
-->

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Interactive Quiz</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>

    <div class="quiz-container">

        <!-- Introduction -->
        <header class="quiz-header">
            <h1>Interactive Quiz</h1>
            <p>Test your knowledge with this fun and interactive quiz!</p>
        </header>

        <!-- Quiz Box -->
        <section id="quiz-box">
            <h2 id="question">Question Text</h2>

            <div class="options">
                <label>
                    <input type="radio" name="option" value="0">
                    <span id="option0">Option 1</span>
                </label>

                <label>
                    <input type="radio" name="option" value="1">
                    <span id="option1">Option 2</span>
                </label>
```

```html
            <span id="option2">Option 3</span>
        </label>

        <label>
            <input type="radio" name="option" value="3">
            <span id="option3">Option 4</span>
        </label>
    </div>

    <button id="next-btn">Next</button>
</section>

<!-- Result Box -->
<section id="result-box" class="hidden">
    <h2>Your Result</h2>
    <p id="score-text"></p>
    <p id="message-text"></p>

    <!-- Feedback Form -->
    <div class="feedback-box">
        <h3>Give Your Feedback</h3>
        <textarea id="feedback-text" placeholder="Write your feedback here..." rows="3"></textarea>
        <button id="submit-feedback-btn">Submit Feedback</button>
        <p id="feedback-msg" class="hidden"></p>
    </div>

    <button id="restart-btn">Restart Quiz</button>
</section>
```

CSS CODE

```css
* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
    font-family: Arial, sans-serif;
}

body {
    background: linear-gradient(to right, #0f2027, #203a43, #2c5364);
    min-height: 100vh;
    display: flex;
    justify-content: center;
    align-items: center;
    color: #333;
}
```

```css
.quiz-container {
    background-color: #ffffff;
    padding: 30px 25px;
    border-radius: 12px;
    width: 90%;
    max-width: 500px;
    box-shadow: 0 10px 25px rgba(0,0,0,0.15);
}

.quiz-header {
    text-align: center;
    margin-bottom: 20px;
}

.quiz-header h1 {
    color: #203a43;
    margin-bottom: 8px;
}

.quiz-header p {
    font-size: 14px;
    color: #555;
}

#question {
    font-size: 20px;
    margin-bottom: 15px;
    color: #203a43;
}

.options label {
    display: block;
    background: #f4f6f8;
    padding: 10px 12px;
    border-radius: 6px;
    margin-bottom: 10px;
```

```css
.options label:hover {
    background: #e8f0f3;
    border-color: #203a43;
}

.options input {
    margin-right: 8px;
}

#next-btn,
#restart-btn {
    display: block;
    width: 100%;
    margin-top: 15px;
    padding: 10px;
    background-color: #ffcc00;
    border: none;
    border-radius: 6px;
    font-size: 15px;
    font-weight: bold;
    cursor: pointer;
    transition: background 0.3s;
}

#next-btn:hover,
#restart-btn:hover {
    background-color: #e6b800;
}

#result-box {
    text-align: center;
}

#result-box h2 {
    color: #203a43;
```

```css
#message-text {
    font-size: 16px;
    font-weight: bold;
    color: #444;
}

.hidden {
    display: none;
}
/* ... your existing CSS remains same ... */

.feedback-box {
    margin-top: 20px;
    padding: 15px;
    border: 1px solid #ddd;
    border-radius: 8px;
    background-color: #f9fafb;
}

.feedback-box h3 {
    margin-bottom: 10px;
    color: #203a43;
}

#feedback-text {
    width: 100%;
    padding: 8px;
    border-radius: 5px;
    border: 1px solid #ccc;
    margin-bottom: 10px;
}

#submit-feedback-btn {
    padding: 8px 16px;
    background-color: #203a43;
    color: white;
    border: none;
```

JavaScript Code

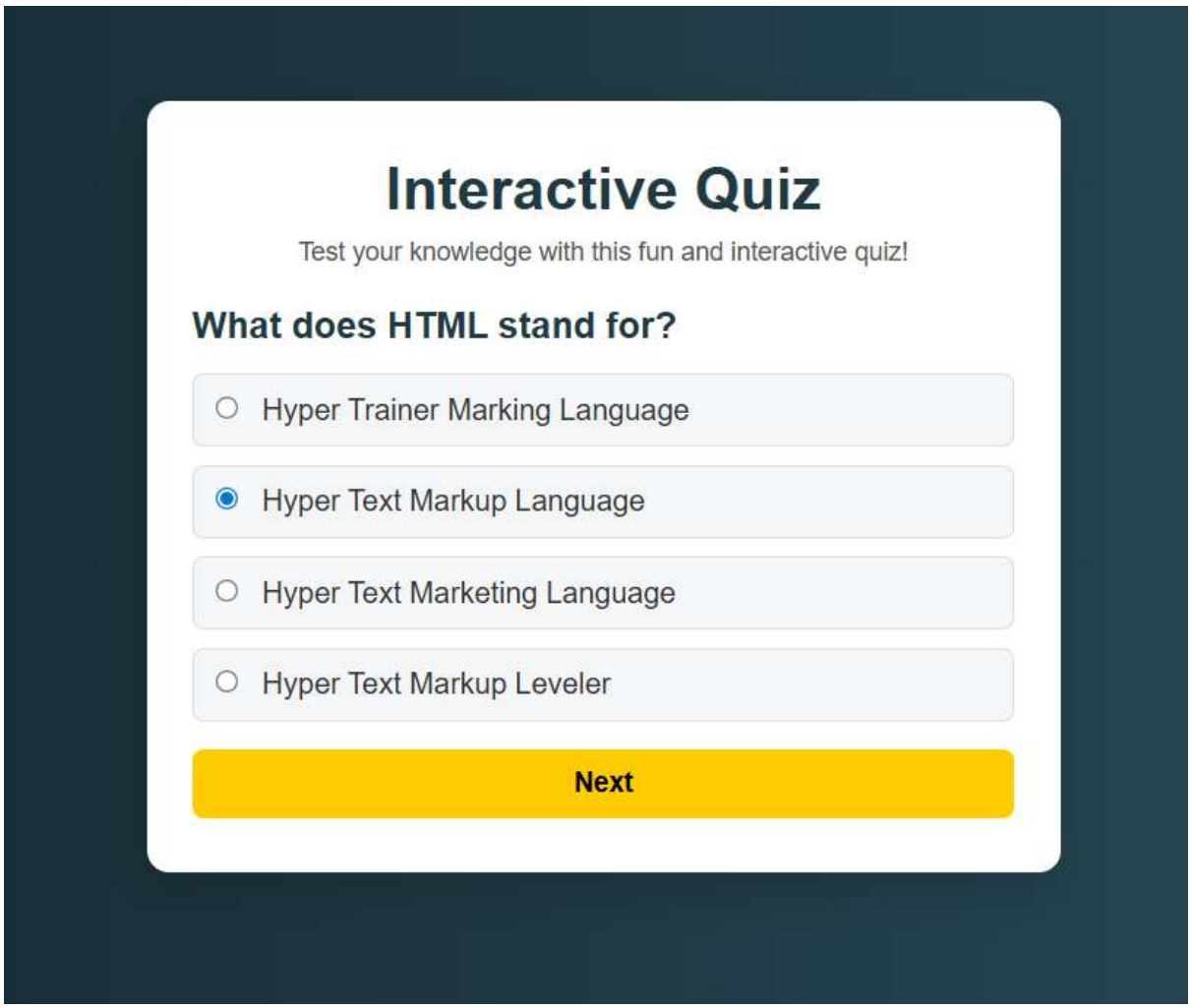
```javascript
const quizData = [
    {
        question: "What does HTML stand for?",
        options: [
            "Hyper Trainer Marking Language",
            "Hyper Text Markup Language",
            "Hyper Text Marketing Language",
            "Hyper Text Markup Leveler"
        ],
        correct: 1
    },
    {
        question: "Which language is used for styling web pages?",
        options: ["HTML", "JQuery", "CSS", "XML"],
        correct: 2
    },
    {
        question: "Which language is used for web interactivity?",
        options: ["Python", "Java", "C++", "JavaScript"],
        correct: 3
    },
    {
        question: "Which tag is used to link CSS to HTML?",
        options: ["<script>", "<style>", "<css>", "<link>"],
        correct: 3
    },
    {
        question: "Which symbol is used for comments in JavaScript?",
        options: ["<!-- -->", "/* */", "//", "#"],
        correct: 2
    }
];
```

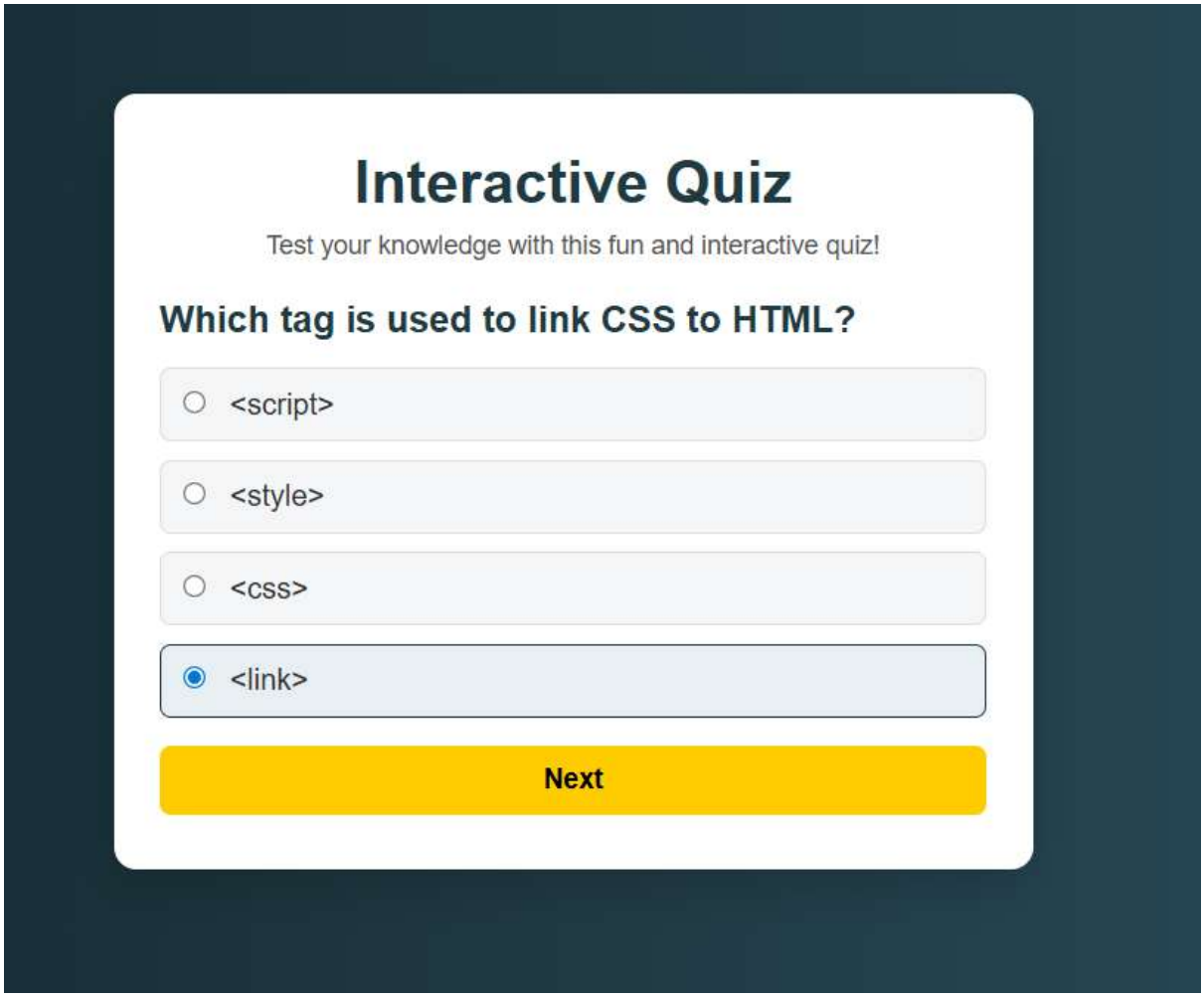
```
const questionEl = document.getElementById("question");
const optionEls = document.getElementsByName("option");
const optionTexts = [
    document.getElementById("option0"),
    document.getElementById("option1"),
    document.getElementById("option2"),
    document.getElementById("option3")
];
const nextBtn = document.getElementById("next-btn");
const quizBox = document.getElementById("quiz-box");
const resultBox = document.getElementById("result-box");
const scoreText = document.getElementById("score-text");
const messageText = document.getElementById("message-text");
const restartBtn = document.getElementById("restart-btn");

// Feedback elements
const feedbackText = document.getElementById("feedback-text");
const submitFeedbackBtn = document.getElementById("submit-feedback-btn");
const feedbackMsg = document.getElementById("feedback-msg");

// Quiz state
let currentQuestionIndex = 0;
let score = 0;

// Load question
function loadQuestion() {
    const currentQuestion = quizData[currentQuestionIndex];
    questionEl.textContent = currentQuestion.question;

    currentQuestion.options.forEach((option, index) => {
        optionTexts[index].textContent = option;
        optionEls[index].checked = false;
    });
}
```

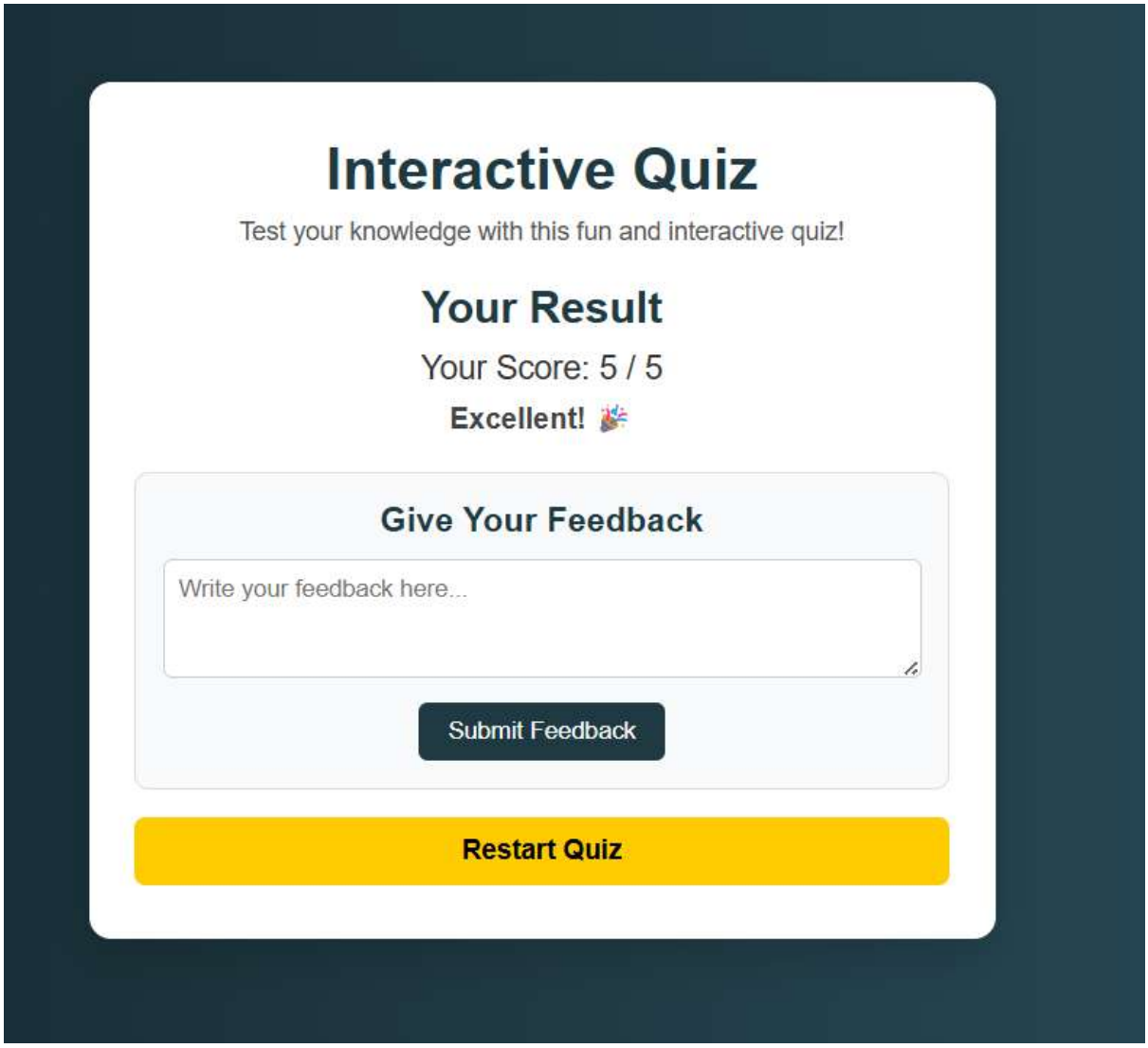
```
function showResult() {
    quizBox.classList.add("hidden");
    resultBox.classList.remove("hidden");

    scoreText.textContent = `Your Score: ${score} / ${quizData.length

    if (score === quizData.length) {
        messageText.textContent = "Excellent! 🎉";
    } else if (score >= quizData.length / 2) {
        messageText.textContent = "Good Job! 👍";
    } else {
        messageText.textContent = "Try Again! 💪";
    }
}

// Next button click
nextBtn.addEventListener("click", () => {
    const selectedAnswer = getSelectedAnswer();

    if (selectedAnswer === null) {
        alert("Please select an option before proceeding!");
        return;
    }

    if (selectedAnswer === quizData[currentQuestionIndex].correct) {
        score++;
    }

    currentQuestionIndex++;

    if (currentQuestionIndex < quizData.length) {
        loadQuestion();
    } else {
        showResult();
    }
});
```

```javascript
// Feedback submit
submitFeedbackBtn.addEventListener("click", () => {
    if (feedbackText.value.trim() === "") {
        alert("Please write some feedback before submitting!");
        return;
    }

    feedbackMsg.textContent = "Thank you for your feedback! 😊";
    feedbackMsg.classList.remove("hidden");
    feedbackText.value = "";
});

// Initial load
loadQuestion();
```

OUTPUT




Interactive Quiz
Test your knowledge with this fun and interactive quiz!
Which tag is used to link CSS to HTML?
<script>
<style>
<css>
<link>
Next



✅ **Result:**

An interactive quiz application was successfully created using HTML, CSS, and JavaScript. The application is user-friendly, visually appealing, and meets all the experiment requirements including scoring, result display, feedback, and restart functionality.

ASSIGNMENT- Dynamic To-Do List

AIM- The aim of this project is to develop a dynamic and interactive To-Do List web application using HTML, CSS, JavaScript, integrated with a Node.js + Express backend and SQLite database. The application allows users to add, edit, delete, mark tasks as completed, filter, search, and sort tasks while ensuring data persistence.

📂 Project Structure-

```
dynamic_todo_list/
|
|— frontend/
|   |— index.html      # HTML structure for UI
|   |— style.css       # Styling for layout, colors, and responsiveness
|   |— script.js       # JavaScript logic for task management and API calls
|
└— backend/
    |— server.js       # Express server handling API requests
    |— package.json    # Node.js project dependencies
    |— db.sqlite       # SQLite database for task persistence
```

Technologies Used

Technology	Purpose
HTML5	Structure of the web interface
CSS3	Layout, styling, priority indicators, hover effects
JavaScript (ES6)	Task management logic, DOM manipulation, and API calls
Node.js	Backend runtime environment
Express.js	Framework for REST API endpoints
SQLite	Lightweight database for persistent storage
CORS	Middleware to allow frontend-backend communication
Fetch API	Connect frontend with backend APIs

📌 Features Implemented

Frontend

- Add Tasks: Users can enter a task title and select priority.
- Edit Tasks: Modify task title or priority.
- Delete Tasks: Remove tasks permanently.
- Mark as Completed: Completed tasks are visually differentiated.
- Filter Tasks: View all, active, or completed tasks.
- Search Tasks: Find tasks by keyword (bonus feature).
- Sort Tasks: Sort by priority level (bonus feature).
- Task Counter: Displays total and completed tasks.

Backend

- **Stores all tasks in SQLite database for persistent storage.**
- **Provides RESTful API endpoints for CRUD operations:**
  - **GET /tasks → Fetch all tasks**
  - **POST /tasks → Add new task**
  - **PUT /tasks/:id → Edit task**
  - **PATCH /tasks/:id/status → Toggle task completion**
  - **DELETE /tasks/:id → Delete task**

💻 Implementation Overview

1. **Frontend**
   - **HTML provides the structure, including input fields, buttons, and task list.**
   - **CSS styles the layout, colors, and responsive behavior.**
   - **JavaScript manages tasks dynamically in the browser, interacting with the backend using fetch API.**
2. **Backend**
   - **Node.js + Express handles API requests.**
   - **SQLite database stores tasks permanently.**
   - **Each API endpoint performs specific CRUD actions and sends responses to the frontend.**
3. **Task Lifecycle**
   - **Tasks are created via the Add button → POST request to backend → database stores task → frontend updates list.**
   - **Tasks can be edited, marked completed, deleted, filtered, searched, and sorted, with changes reflected in both UI and database.**

CODE – Frontend

Index.html

```html
<!DOCTYPE html>
<html>
<head>
<title>My To-Do List</title>
<link rel="stylesheet" href="style.css">
</head>

<body>

<h1>My To-Do List</h1>

<div class="input-box">

<input type="text" id="taskTitle" placeholder="Enter task title">

<select id="priority">
<option value="Low">Low</option>
<option value="Medium" selected>Medium</option>
<option value="High">High</option>
</select>

<button onclick="addTask()">Add Task</button>

</div>

<div class="bonus">

<input type="text" id="searchBox" placeholder="Search task..." onkeyup="searchTask()">

<button onclick="sortPriority()">Sort by Priority</button>

</div>

<div class="filters">
```

```html
<button onclick="filterTasks('all')">All</button>
<button onclick="filterTasks('active')">Active</button>
<button onclick="filterTasks('completed')">Completed</button>

</div>

<ul id="taskList"></ul>

<p id="counter"></p>

<script src="script.js"></script>
```

Style.css

```
body {
    background: linear-gradient(135deg, #ffe6e6, #fff0f5);
    text-align: center;
    margin: 0;
    padding: 0;
}

/* Heading styling */
h1 {
    margin-top: 30px;
    font-size: 2em;
    color: #333;
}

/* Input and select styling */
input, select {
    padding: 10px 12px;
    margin: 8px 5px;
    border: 1px solid #ccc;
    border-radius: 5px;
    font-size: 1em;
    outline: none;
    transition: 0.3s;
}

input:focus, select:focus {
    border-color: #007bff;
    box-shadow: 0 0 5px rgba(0,123,255,0.3);
}

/* Button styling */
button {
    padding: 10px 16px;
    margin: 8px 5px;
    cursor: pointer;
    border: none;
    border-radius: 5px;
```

```
/* List styling */
ul {
    padding: 0;
    margin: 20px auto;
    max-width: 450px;
}

li {
    list-style: none;
    background: white;
    margin: 10px 0;
    padding: 12px 15px;
    border-radius: 8px;
    display: flex;
    justify-content: space-between;
    align-items: center;
    box-shadow: 0 2px 6px rgba(0,0,0,0.1);
    transition: 0.3s;
}

li:hover {
    transform: translateY(-2px);
    box-shadow: 0 4px 10px rgba(0,0,0,0.15);
}

/* Completed task styling */
.completed {
    text-decoration: line-through;
    color: gray;
    opacity: 0.7;
}

/* Priority colors */
.high {
    color: #e74c3c; /* bright red */
    font-weight: bold;
}
```

```css
.medium {
    color: #f39c12; /* orange */
    font-weight: bold;
}

.low {
    color: #2ecc71; /* green */
    font-weight: bold;
}
```

Script.js

```js
let tasks=[]

function loadTasks(){

fetch("http://localhost:3000/tasks")
.then(res=>res.json())
.then(data=>{
tasks=data
displayTasks(tasks)
})

}

function addTask(){

let title=document.getElementById("taskTitle").value
let priority=document.getElementById("priority").value

if(title=="" || title.length>50){
alert("Task title invalid")
return
}
```

```
function displayTasks(list){
    let ul=document.getElementById("taskList")
    ul.innerHTML=""

    list.forEach(task=>{
        let li=document.createElement("li")

        if(task.isDone){
            li.classList.add("completed")
        }

        li.innerHTML=`
        <span class="${task.priority.toLowerCase()} task-text">
            ${task.title} (${task.priority})
        </span>

        <div>
            <button onclick="toggleTask(${task.id})">✓</button>
            <button onclick="editTask(${task.id}, '${task.title}', '${task.priority}')">✏️</button>
            <button onclick="deleteTask(${task.id})">❌</button>
        </div>
        `

        ul.appendChild(li)
    })

    updateCounter()
}
function editTask(id, currentTitle, currentPriority){

    let newTitle = prompt("Edit Task Title:", currentTitle)
    if(!newTitle || newTitle.trim()=="" || newTitle.length>50) return

    let newPriority = prompt("Edit Priority (Low / Medium / High):", currentPriority)
    if(!["Low","Medium","High"].includes(newPriority)) return
```

```
function toggleTask(id){

fetch(`http://localhost:3000/tasks/${id}/status`,{
method:"PATCH"
})
.then(()=>loadTasks())

}

function deleteTask(id){
```

```
function deleteTask(id){

fetch(`http://localhost:3000/tasks/${id}`,{
method:"DELETE"
})
.then(()=>loadTasks())

}

function filterTasks(type){

if(type=="all")
displayTasks(tasks)

if(type=="active")
displayTasks(tasks.filter(t=>t.isDone==0))

if(type=="completed")
displayTasks(tasks.filter(t=>t.isDone==1))

}

function updateCounter(){

let total=tasks.length
let completed=tasks.filter(t=>t.isDone==1).length

document.getElementById("counter").innerText=
`Completed ${completed} / ${total}`

}

/* BONUS FEATURE 1 */
function searchTask(){

let keyword=document.getElementById("searchBox").value.toLowerCase()
```

```
let filtered=tasks.filter(task =>
task.title.toLowerCase().includes(keyword)
)

displayTasks(filtered)

}

/* BONUS FEATURE 2 */
function sortPriority(){

let order={
High:1,
Medium:2,
Low:3
}

let sorted=[...tasks].sort((a,b)=>{
return order[a.priority]-order[b.priority]
})

displayTasks(sorted)

}

loadTasks()
```

Backend – server.js

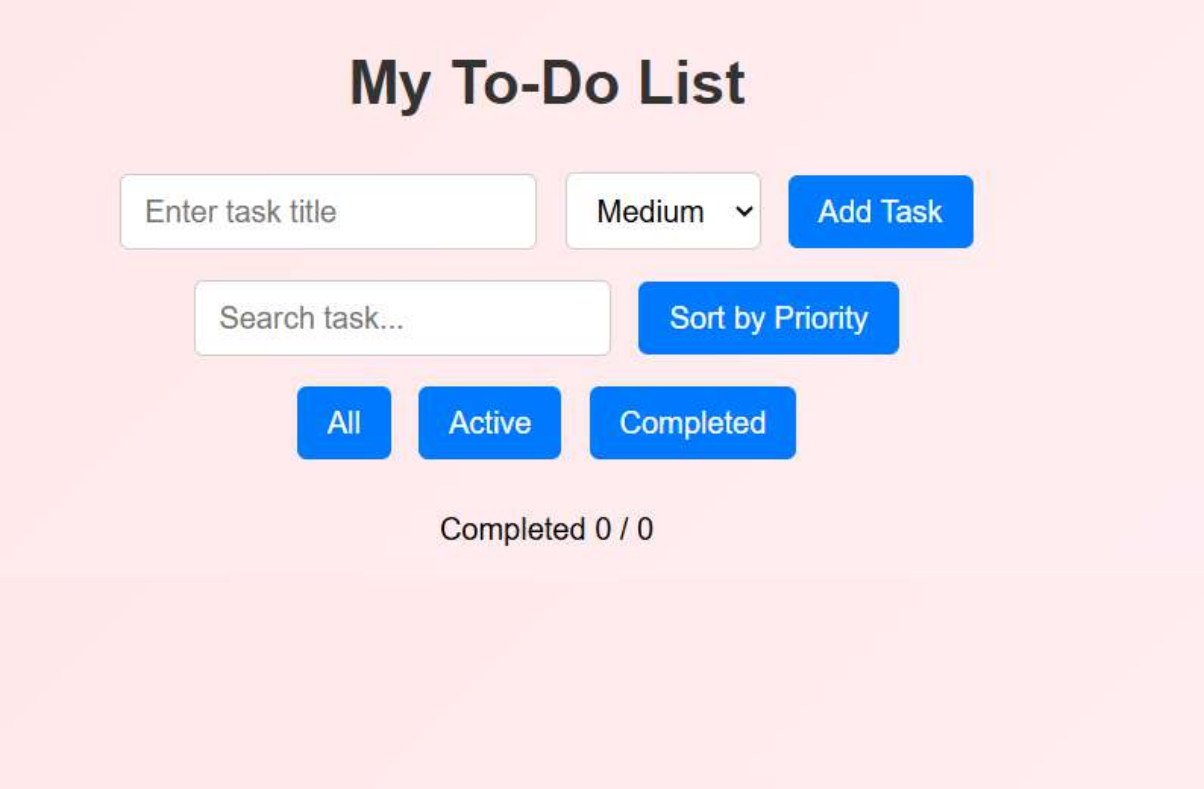
```
const express=require("express")
const sqlite3=require("sqlite3").verbose()
const cors=require("cors")

const app=express()

app.use(cors())
app.use(express.json())

const db=new sqlite3.Database("db.sqlite")

db.run(`
CREATE TABLE IF NOT EXISTS tasks(
id INTEGER PRIMARY KEY AUTOINCREMENT,
title TEXT NOT NULL,
priority TEXT DEFAULT 'Medium',
isDone INTEGER DEFAULT 0,
createdAt DATETIME DEFAULT CURRENT_TIMESTAMP
)
`)

app.get("/tasks",(req,res)=>{
db.all("SELECT * FROM tasks",(err,rows)=>{
res.json(rows)
})
})

app.post("/tasks",(req,res)=>{
const {title,priority}=req.body

db.run(
"INSERT INTO tasks(title,priority) VALUES (?,?)",
[title,priority],
function(){
res.json({id:this.lastID})
})
})
```

```
app.put("/tasks/:id",(req,res)=>{
const {title,priority}=req.body

db.run(
"UPDATE tasks SET title=?,priority=? WHERE id=?",
[title,priority,req.params.id],
()=>res.json({message:"updated"})
)
})

app.patch("/tasks/:id/status",(req,res)=>{
db.run(
"UPDATE tasks SET isDone = NOT isDone WHERE id=?",
[req.params.id],
()=>res.json({message:"status changed"})
)
})

app.delete("/tasks/:id",(req,res)=>{
db.run(
"DELETE FROM tasks WHERE id=?",
[req.params.id],
()=>res.json({message:"deleted"})
)
})

app.listen(3000,()=>{
console.log("Server running on port 3000")
})
```

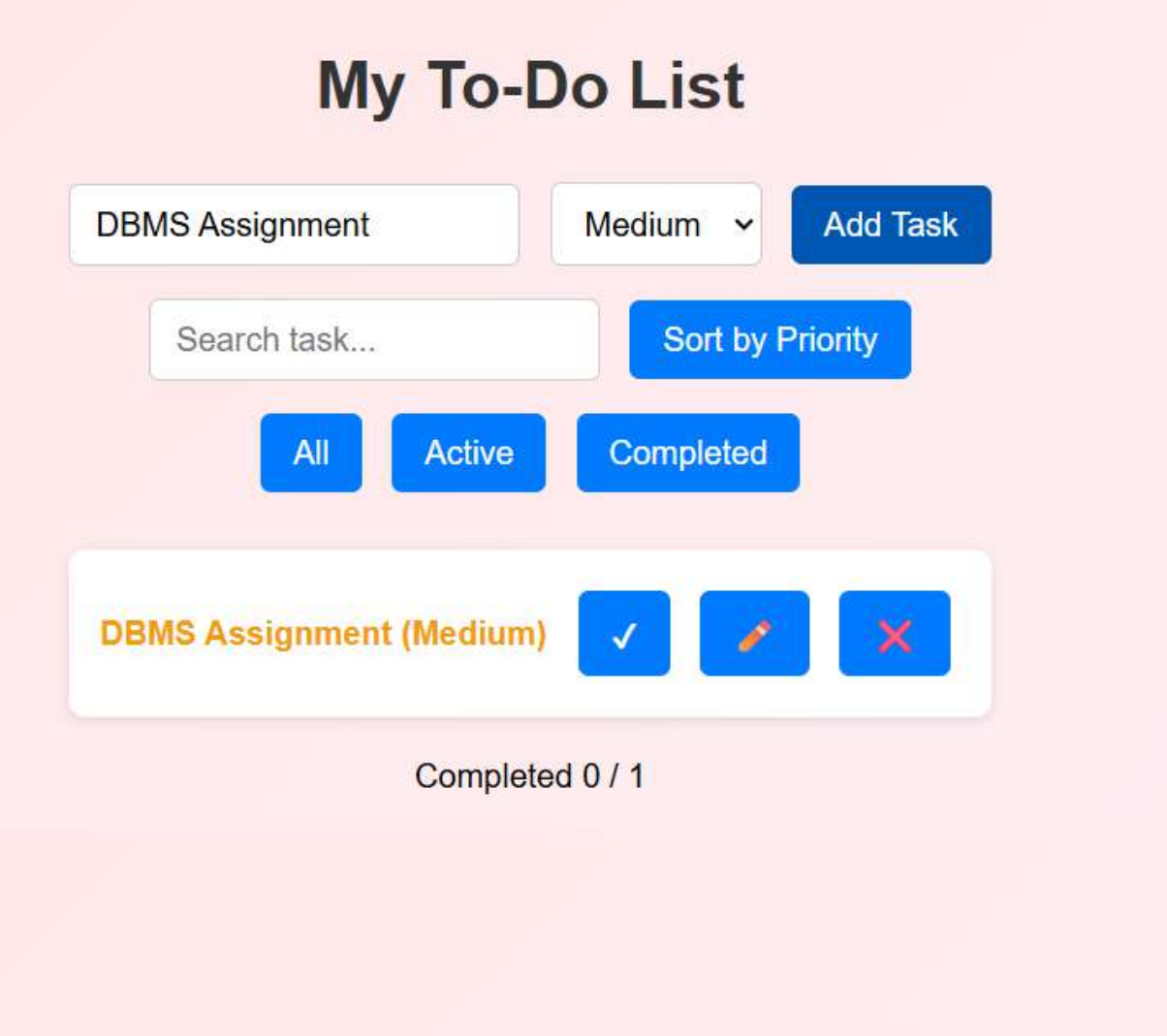
Package.json

```
{
"name":"todo-backend",
"version":"1.0.0",
"main":"server.js",
▷Debug
"scripts":{
"start":"node server.js"
},
"dependencies":{
"express":"^4.18.2",
"sqlite3":"^5.1.6",
"cors":"^2.8.5"
}
}
```

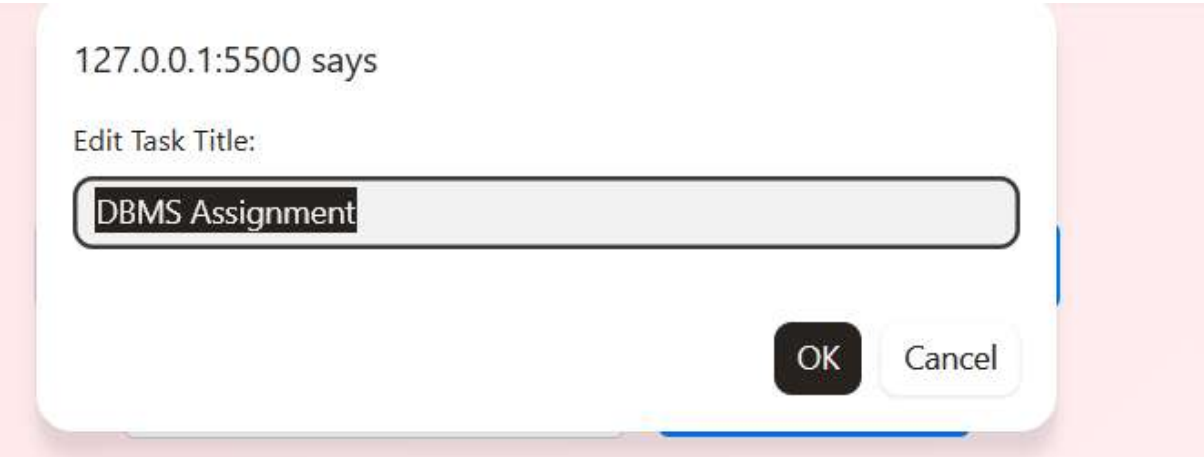
OUTPUT

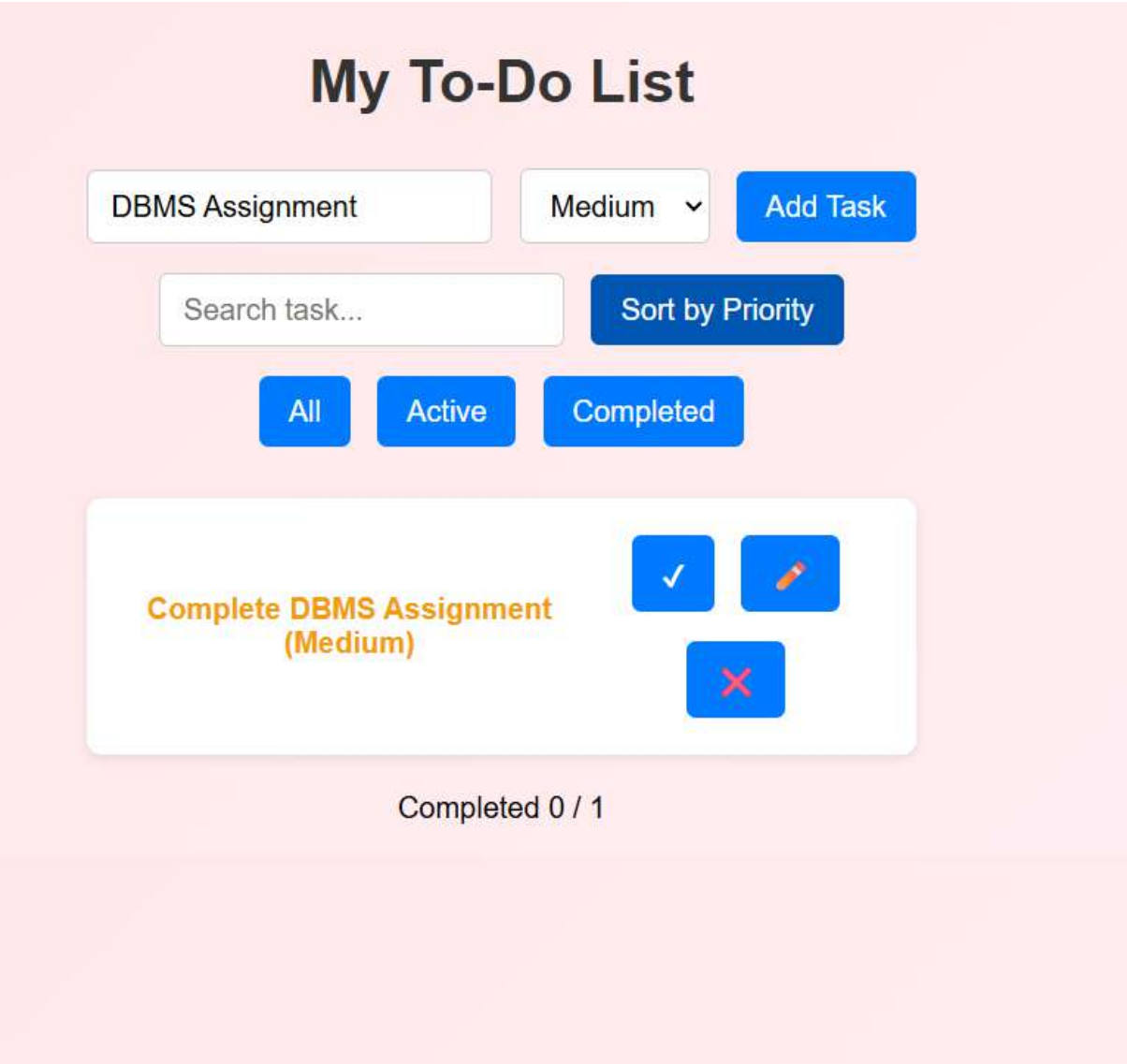


➕ Add Task

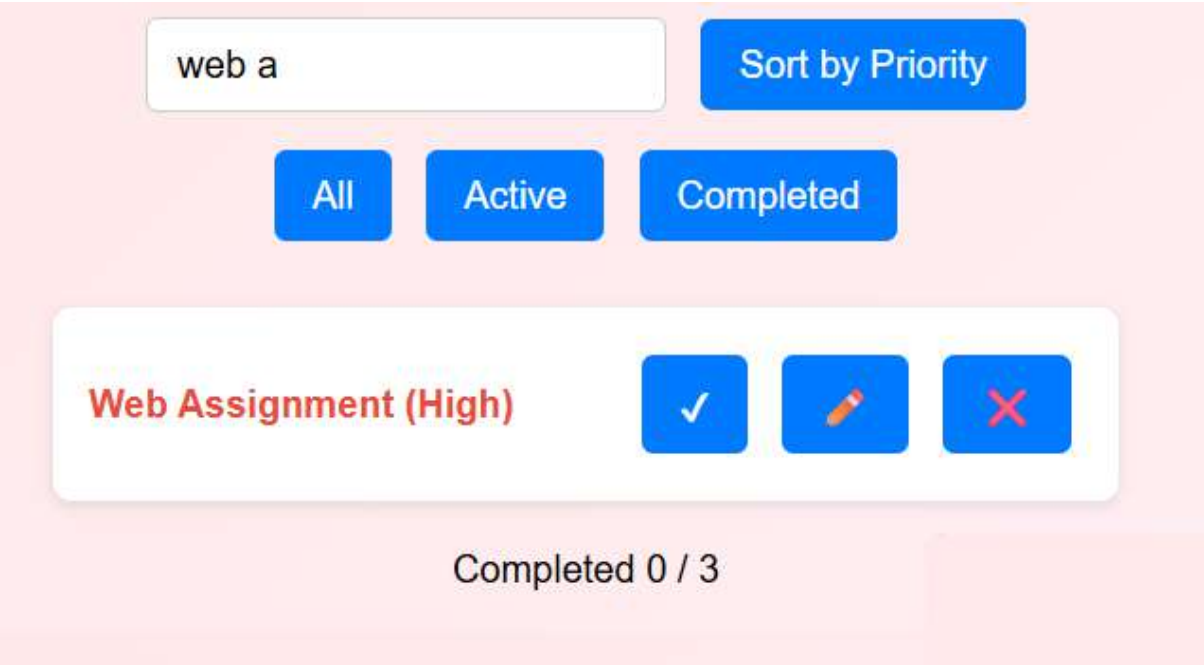


🖊️ Edit Task

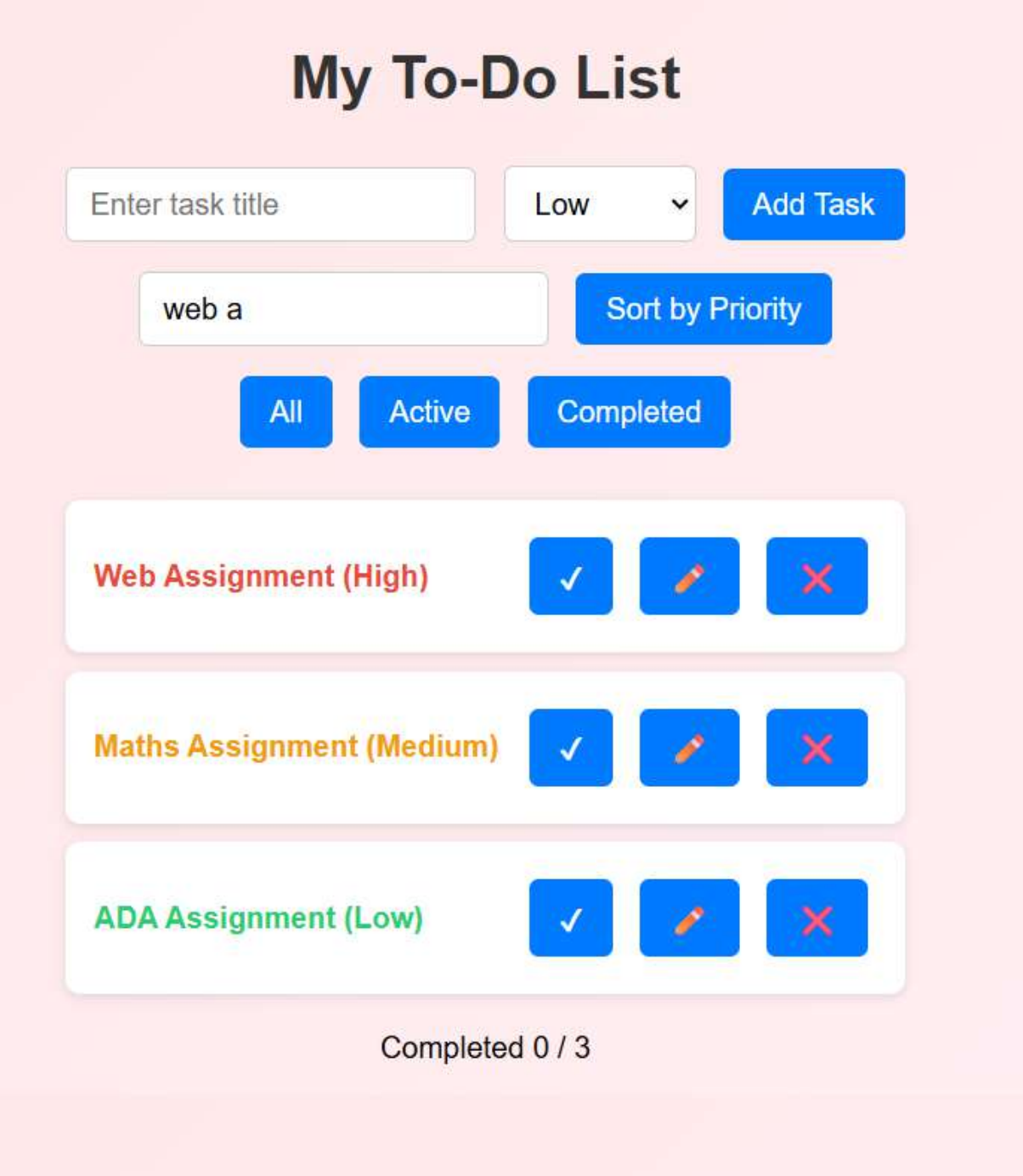



My To-Do List
DBMS Assignment
Medium
Add Task
Search task...
Sort by Priority
All
Active
Completed
Complete DBMS Assignment (Medium)
Completed 0 / 1

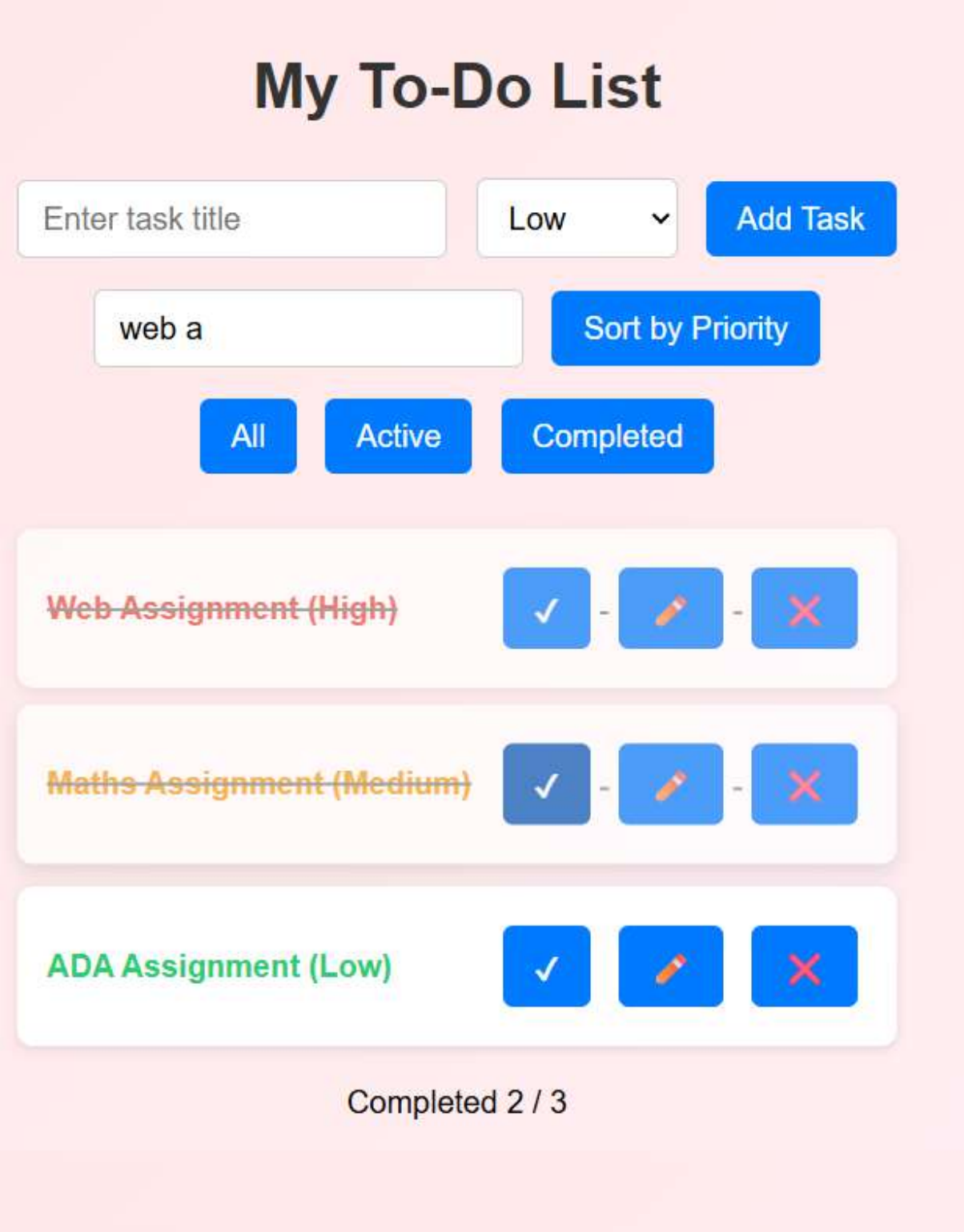
🔎 **Search by Keyword**



📝 **Sort by Priority**



✅ **Complete Task**



Conclusion

The Dynamic To-Do List application demonstrates full-stack development by integrating frontend interactivity with backend persistence.
Students learn practical skills in HTML, CSS, JavaScript, Node.js, Express, and SQLite while implementing a real-world web application with CRUD functionality.

🎯 Learning Outcomes

By completing this project, students will be able to:

- **Design a responsive UI using HTML and CSS.**
- **Implement DOM-based CRUD operations using JavaScript.**
- **Use REST API endpoints to perform CRUD operations with a database.**
- **Validate user input and handle edge cases (e.g., empty or too long task titles).**
- **Understand the difference between in-memory storage and persistent storage.**
- **Implement bonus features like search and sort.**